

Program 3: Caesar Cipher Cryptosystem Demonstration

Code:

```
# Program 3: Caesar Cipher Cryptosystem Demonstration
# Reads plaintext from a file and writes ciphertext to another file

def caesar_encrypt(text, shift):
    result = ""
    for ch in text:
        if ch.isalpha():
            base = 'A' if ch.isupper() else 'a'
            result += chr((ord(ch) - ord(base) + shift) % 26 + ord(base))
        else:
            result += ch
    return result

# File paths
input_file = "input_plaintext.txt"
output_file = "caesar_output.txt"
shift = int(input("Enter shift value for Caesar Cipher: "))

# Read plaintext
with open(input_file, "r") as f:
    plaintext = f.read()

# Encrypt
ciphertext = caesar_encrypt(plaintext, shift)
```

```
# Write ciphertext
with open(output_file, "w") as f:
    f.write(ciphertext)
print("Caesar Cipher Encryption Completed!")
```

Output:

Enter shift value for Caesar Cipher: 3
Caesar Cipher Encryption Completed!

PLAIN TEXT: My Name is Meet Shah.

CIPHER TEXT: pb qdph lv phhw vkdk.

Program4: Monoalphabetic Cipher Cryptosystem Demonstration

Code:

```
# Program 4: Monoalphabetic Cipher Cryptosystem Demonstration
# Reads plaintext from a file and writes ciphertext to another file

# Example key mapping (A-Z)
key = "QWERTYUIOPASDFGHJKLZXCVBNM"

def mono_encrypt(text):
    result = ""
    for ch in text:
        if ch.isalpha():
            if ch.isupper():
```

```

        result += key[ord(ch) - ord('A')]
    else:
        result += key[ord(ch) - ord('a')].lower()
    else:
        result += ch
    return result

input_file = "input_plaintext.txt"
output_file = "mono_output.txt"

# Read plaintext
with open(input_file, "r") as f:
    plaintext = f.read()

# Encrypt
ciphertext = mono_encrypt(plaintext)

# Write ciphertext
with open(output_file, "w") as f:
    f.write(ciphertext)
print("Monoalphabetic Cipher Encryption Completed!")

```

Output:

Monoalphabetic Cipher Encryption Completed!

PLAIN TEXT: My Name is Meet Shah.

CIPHER TEXT: dn fqdt ol dttz liqi.

Program5: Write a program to demonstrate the Vigenère Cipher

Code:

```
# Program 5: Vigenère Cipher Demonstration

# Input from file and output to file

def vigenere_encrypt(plaintext, key):
    result = ""
    key = key.upper()
    key_index = 0
    for ch in plaintext:
        if ch.isalpha():
            shift = ord(key[key_index % len(key)]) - ord('A')
            base = 'A' if ch.isupper() else 'a'
            result += chr((ord(ch) - ord(base) + shift) % 26 + ord(base))
            key_index += 1
        else:
            result += ch
    return result

input_file = "input_plaintext.txt"
output_file = "vigenere_output.txt"
key = input("Enter Vigenere key: ")
with open(input_file, "r") as f:
    plaintext = f.read()
```

```
ciphertext = vigenere_encrypt(plaintext, key)
with open(output_file, "w") as f:
    f.write(ciphertext)
print("Vigenere Cipher Encryption Completed!")
```

Output:

Enter Vigenere key: encrypt

Vigenere Cipher Encryption Completed!

PLAIN TEXT: My Name is Meet Shah.

CIPHER TEXT: qi prkt bw zgvr haeu.

Program 6: Write a Program to demonstrate the AutoKey Cipher

Code:

```
# Program 6: AutoKey Cipher Demonstration
```

```
# Input from file and output to file
```

```
def autokey_encrypt(plaintext, key):
```

```
    result = ""
```

```
    full_key = key + plaintext # AutoKey = key + plaintext characters
```

```
    key_index = 0
```

```
    for ch in plaintext:
```

```
        if ch.isalpha():
```

```
    shift = (ord(full_key[key_index].upper()) - ord('A'))
    base = 'A' if ch.isupper() else 'a'
    result += chr((ord(ch) - ord(base) + shift) % 26 + ord(base))
    key_index += 1
else:
    result += ch
return result
```

```
input_file = "input_plaintext.txt"
output_file = "autokey_output.txt"
```

```
key = input("Enter AutoKey initial key: ")
```

```
with open(input_file, "r") as f:
    plaintext = f.read()
ciphertext = autokey_encrypt(plaintext, key)
with open(output_file, "w") as f:
    f.write(ciphertext)
print("AutoKey Cipher Encryption Completed!")
```

Output:

PLAINTEXT: My Name is Meet Shah.

Enter AutoKey initial key: Encrpyt

CIPHER TEXT: qi prbc be kxrt eltp.